

JoTouch Real Training Results (main model)

- Model: JoTouchModel (MobileNetV2 + TCN) — MobileNetV2 confirmed over MobileNetV3 by the upgrade evaluation (see `model_comparison_results.md`)
- Pipeline (full): AdamW + OneCycleLR + gradient clipping + early stopping (patience 8), up to 40 epochs, batch 64, lr=0.001
- Split per dataset: 70% train / 15% val / 15% untouched test
- Device: cpu

Dataset	Classes	Epochs trained	Test Accuracy	Latency (ms)	Params	Saved weights
Zenodo MDPI FMG (17 subjects, 5 motions)	5 (1D_X, 1D_Y, 2D_DG, 2D_DM, 2D_SQ)	13	86.78%	3.57	65,797	jotouch_real_zenodo.pt
Exoskelebox FMG benchmark (20 subjects, 6 gestures)	6 (closed, extension, flexion, rest, straight, wide)	40	70.92%	3.07	65,862	jotouch_real_exoskelebox.pt
PLoS ONE 2025 FMG (27 participants, 4 gestures)	4 (Key, Pinch, Power, Tripod)	12	42.61%	2.06	65,732	jotouch_real_plos.pt

Notes

- One model per dataset: the label sets have different meanings across datasets, so merging them into one classifier would be dishonest.
- Input: 8 FMG channels x 20 timesteps (200 ms), min-max normalized per dataset (the loaders set `NUM_SENSORS=8` to match the open data; the on-device default of 4 sensors in `JoTouch_AI_module.py` is unchanged).
- These test accuracies are in-distribution numbers. The cross-subject (new user) estimates remain the LOSO figures in `model_comparison_results.md`.